

Per-stage runtime of the four-index BIE pipeline on the graphene-monolayer system (precompute-once volume field)

Numerical experiment 6.5 — exp65_orbital_bench

2026-06-15

1 Setup

Single right-hand-side solve of the screened-Coulomb four-index integral on the real graphene monolayer (Wannier p_z orbital, k_{323201}), using the production pipeline with the `PrecomputedVolumeField` (`cache_fft`) volume-field method for the volume-potential stages. All timings are warm (post-JIT), measured on a single idle node.

Hardware / software	
CPU	2× AMD EPYC 9474F (96 cores), 96 threads
Memory	1.50 TB (peak RSS 29.40 GB)
OS / kernel	Rocky Linux 8.10, kernel 6.6
Julia	1.12.6
Libraries	FINUFFT.jl 3.5.2, finufft_jll 2.5.1, FMM3D
Problem	
Geometry	slab $L \times L \times L_z = 90 \times 90 \times 3.35$, $\varepsilon_{\text{in}} = 3.5$
Source	squared p_z orbital, 355 862.00 grid points
Tolerances	source/RHS 10^{-3} , LHS/eval 10^{-5} , GMRES 10^{-5}
Discretization	$n_{\text{quad}} = 6$, edge level 4, edge correction on
Interface	960 768.00 points; GMRES 8 iters, residual 7.10×10^{-6}
Result	$u_{\text{onsite}} = 9.88 \text{ eV}$

2 Per-stage runtime

3 Notes

- **Dominant costs** are GMRES (22.20 s), LHS assembly (13.60 s), field construction (8.80 s), and the interface build (6.80 s); the pipeline is solve/assembly-bound.
- **Amortization.** The RHS-independent precompute (field construction + interface build + LHS assembly = 29.10 s) is paid once. Each additional right-hand side on the same geometry/source family costs RHS + GMRES + eval ≈ 30.00 s, dominated by GMRES.
- **Volume field.** The precomputed spectral field is reused across all mesh-refinement depths and the evaluation; with `cache_fft` the u_{int} read-out is interpolation-only (0.12 s).

Phase	Stage	Time (s)
Load	XSF read + truncation + screening	3.25
Precompute	Field construction (<code>cache_fft</code>)	8.82
Precompute	Interface build (RHS-adaptive)	6.76
Precompute	LHS assembly (near corrections)	13.56
Solve	RHS assembly (field)	2.18
Solve	GMRES solve (8 iterations)	22.21
Eval	Volume potential u_{int} (field)	0.12
Eval	Scattered potential (FMM + cubature)	5.38
Total		62.28

Table 1: Wall-clock time per pipeline stage (warm, 96 threads).

Phase	Time (s)	Share (%)
Load	3.25	5.20
Precompute (construct + build + LHS)	29.14	46.80
Solve (RHS + GMRES)	24.39	39.20
Eval (u_{int} + scatter)	5.50	8.80
Total	62.28	100.00

Table 2: Phase totals.

- **Memory.** Peak RSS 29.40 GB, set by the `cache_fft` fine-grid storage; the standard field (`cache_fft = false`) peaks at ~ 14.00 GB with $u_{\text{int}} \approx 0.40$ s.

4 Multiple right-hand sides

The four-index matrix is assembled by solving many right-hand sides on *one* shared interface. We benchmark a central graphene p_z orbital together with its lattice neighbours (the `lattice_scale` demo convention): K pair densities $\rho_{1j} = \phi_1 \phi_j$ (on-site plus neighbour shells out to a cutoff) are solved together as a batched $n_d=K$ FMM operator with block GMRES, and the central row of the Coulomb matrix, $V[\rho_{11}, \rho_j]$, is evaluated on the fixed on-site target set. The neighbour cutoff controls K . To run at the full 96 threads we use the **ducc0 FFT backend** for FINUFFT: the stock FFTW build has a $\sim 50 \times$ 96-thread plan pathology on these upsampled mode grids, which cripples *both* the interface build and the evaluation. Each cutoff runs as its own Slurm-array task on a dedicated 96-core genoa node (2026-06-17).

- **Fixed setup amortizes.** Precompute (interface + batched LHS, ≈ 37.00 s) and the eval-side `pottrg` map build (≈ 58.00 s, the corrected layer-potential map over the fixed on-site target) are K -independent — both are paid once per batch and spread over all K right-hand sides. (The $K=4$ `pottrg` value was confirmed against a slower-node outlier by a re-run on a normal node, 55.60 s.)
- **Block solve is sublinear** in K ($\sim K^{0.85}$): the batched $n_d=K$ FMM amortizes the tree build and near-field setup over the columns, so each added right-hand side is cheaper than the first

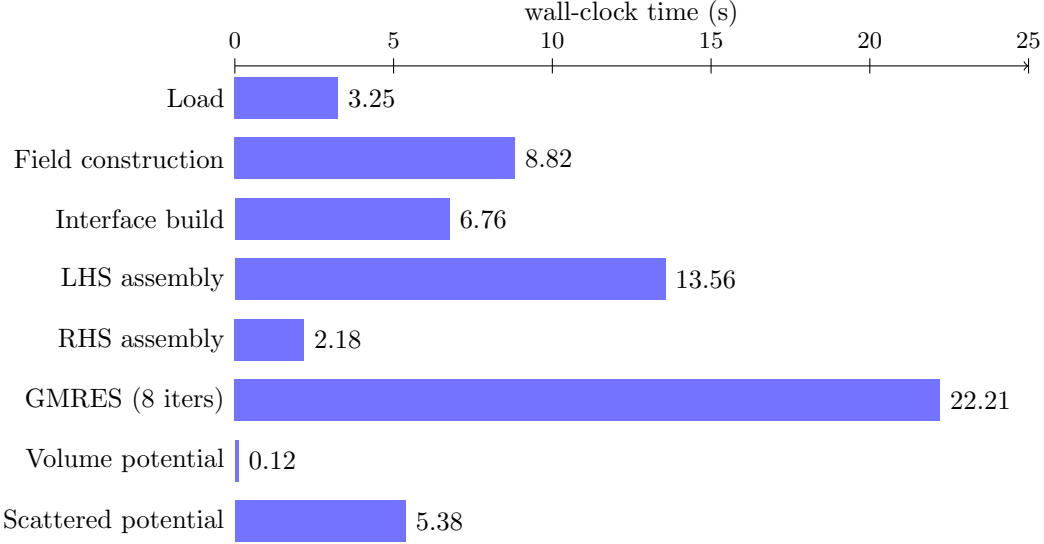


Figure 1: Per-stage wall-clock time. The pipeline is dominated by GMRES, LHS assembly, and the one-time field construction; the volume-potential evaluation (u_{int}) is essentially free.

K	Precompute (s)	pottrg build (s)	Block solve (s)	Eval (s)	Total (s)	Peak RAM (GB)
1	35.2	63.6	20.6	2.6	125.7	16.6
4	34.5	55.6	30.1	11.7	136.1	40.1
10	38.5	59.0	65.4	35.8	206.6	89.0
13	36.4	59.1	79.3	43.6	230.7	113.4
19	39.4	53.4	112.0	66.2	285.0	163.4

Table 3: Per-phase wall-clock time and peak RAM versus the number of right-hand sides K (graphene central orbital + neighbour shells; cutoffs 0, 1.5, 2.5, 2.9, 3.9 bohr $\rightarrow K = 1, 4, 10, 13, 19$). Warm, ducc0 backend, 96 threads. *Precompute* = envelope + interface build + batched LHS; *pottrg build* = corrected layer-potential map onto the fixed on-site target; *Block solve* = RHS assembly + block GMRES; *Eval* = the K on-site-row field evaluations.

(20.60 s at $K=1$ vs. ≈ 5.50 s marginal at large K).

- **Eval is linear** in K (≈ 3.50 s per pair): one TKM volume potential plus one corrected layer-potential apply per right-hand side, all on the same fixed on-site target set.
- **Peak RAM grows linearly** ($\approx 18 + 7.5 K$ GB), set by the block-GMRES $n_d=K$ FMM workspace; 163.00 GB at $K=19$ on the 1.50 TB node.
- **ducc0 vs. FFTW**. Switching FINUFFT to the ducc0 FFT backend removes the FFTW 96-thread cliff (interface build at cutoff 2.5: 26.00 s with ducc0 vs. 171.00 s with FFTW, both at 96 threads), letting the whole pipeline run at full 96 threads. $u_{\text{onsite}} \approx 7.10$ eV across K (midplane orbital convention; cf. the 9.88 eV single-RHS value above, which uses the top-face convention).

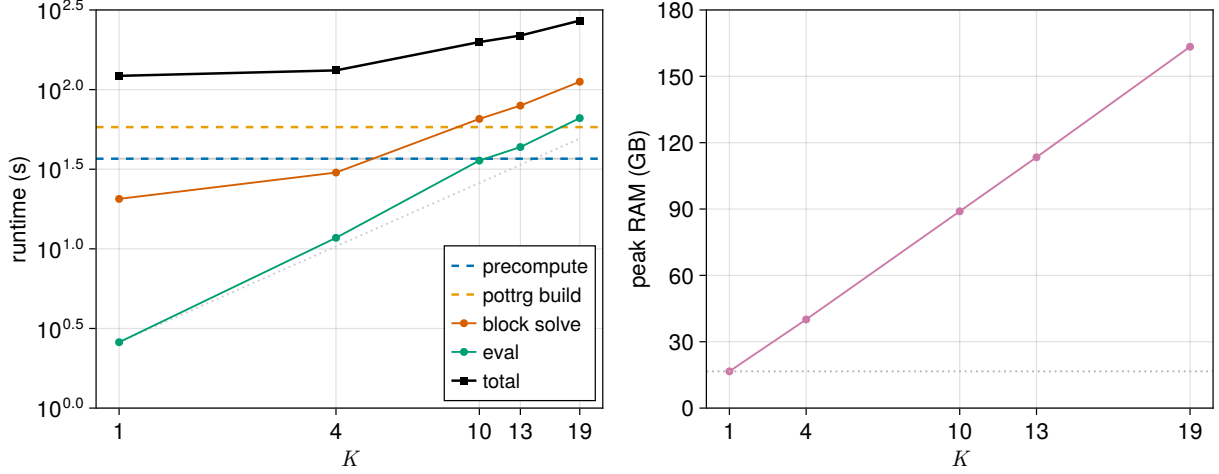


Figure 2: Per-phase runtime (left) and peak RAM (right) versus K . The RHS-independent setup costs (precompute, pottrg build) are flat; the block solve is sublinear and the eval is linear in K ; peak RAM grows linearly.

5 Multiple right-hand sides on a heterojunction substrate (exp66)

The same multi-RHS pipeline applied to a multi-dielectric *heterojunction* substrate: two $90 \times 90 \times 90$ cubes of different permittivity (Si/SiO₂, $\epsilon = 11.9 | 3.9$) sharing the buried internal face $x = c_x$, with a $90 \times 90 \times 9$ slab ($\epsilon = 10$) on top, anchored on the central graphene p_z orbital (slab centre = orbital centroid, junction directly beneath); see Fig. 3. The orbital plus its lattice neighbours within a cutoff form K pair densities $\rho_{1j} = \phi_1 \phi_j$, block-solved on one shared interface, and the central row $V[\rho_{11}, \rho_{1j}]$ is evaluated. The only pipeline change from the single-slab case above is **box-based multi-region screening** (the interface spans three ϵ regions, so the single-region interface screening cannot be applied); 96 threads run via the TKM3D in-function FINUFFT nthreads cap (no DUCC override needed here).

K	Precompute (s)	Block solve (s)	Eval (s)	Total (s)	Peak RAM (GB)
1	7.3	195.2	5.4	215.4	28.4
4	12.0	402.7	27.4	454.4	75.7
10	10.2	624.8	53.5	709.8	176.8
13	11.3	778.9	67.5	884.7	226.1
19	8.1	1074.0	90.6	1208.1	328.9

Table 4: exp66 heterojunction: per-phase wall-clock time and peak RAM versus the number of right-hand sides K (cutoffs 0, 1.5, 2.5, 2.9, 3.9 bohr $\rightarrow K = 1, 4, 10, 13, 19$; 96 threads). The interface is 1617264.00 points, K -independent. The pottrg build is omitted (≈ 0.20 s, negligible — see below); *Total* is the full warm-pipeline wall (it also includes a bare-reference TKM, so it slightly exceeds the column sum).

- **Onsite self-energy.** $V[\rho_{11}, \rho_{11}] \approx 70.4$ (raw kernel units) against the bare interface-free value $V_{\text{vac}} \approx 590$ — the environment screens it $\approx 8.4\times$. $V[11]$ is K -independent to $\sim 0.2\%$

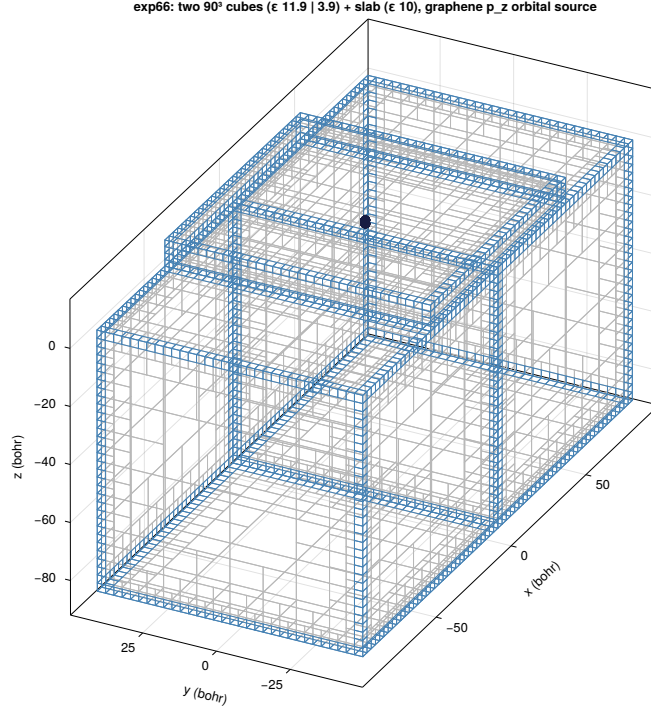


Figure 3: The exp66 multicube system rendered with BL.jl’s Makie extension (`viz_3d`): two 90° cubes ($\epsilon = 11.9 | 3.9$) sharing the buried face, the slab ($\epsilon = 10$) on top, and the graphene p_z orbital density (dark cluster) at the slab centre over the junction. Coarse interface, visualization only.

(it does not depend on the neighbours) and matches the single-RHS production value ($70.30, 8.40\times$).

- **Block solve dominates**, sublinear per RHS ($195 \rightarrow 57$ s/RHS); **eval is linear** (≈ 5 s/pair); **precompute is K -independent** (interface = 1.62M points).
- **pottrg is negligible** (≈ 0.20 s, vs. ≈ 55.00 s for the graphene slab above): the thick slab keeps the orbital clear of the interface, so the target-panel neighbour list is *empty* (0 hcubature corrections vs. 12,724 there) and the corrected layer-potential map collapses to its smooth FMM far field.
- **Conditioning**. The high-contrast three-region junction needs 30–36 block-GMRES iterations (vs. ~ 8 for the single slab), and the ~ 1.60 M-point interface drives peak RAM to 329.00 GB at $K=19$ (≈ 16 GB/ K).

6 Full lattice campaign: 10×10 graphene on the heterojunction

Production four-index run for the full **200-orbital** 10×10 graphene supercell on the exp66 multicube substrate (‘lattice_scale’ campaign `lattice_10x10_multicube`). The pairing cutoff (5.00 lattice units) gives up to 30 neighbours per bulk centre (6 shells), tapering to 8 at the flake corners, for **2611 distinct pair densities** ρ_{ij} — the basis of the 2611×2611 Coulomb tensor $V_{ij,kl}$. Run as a

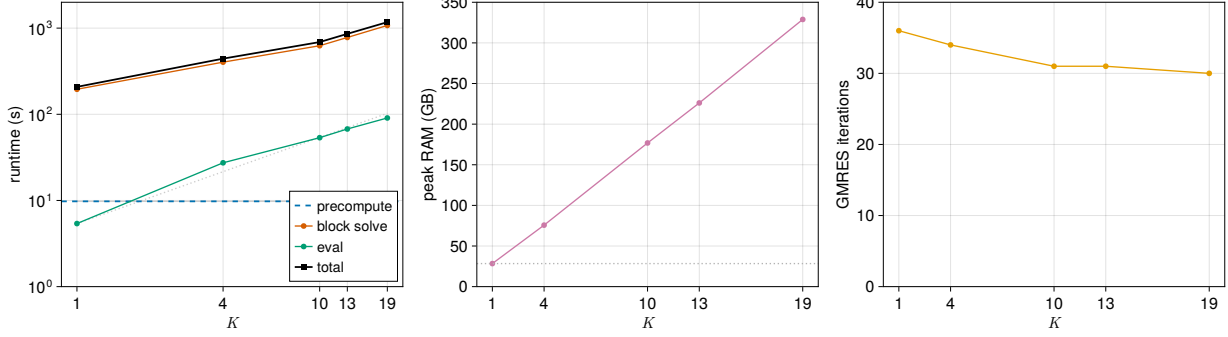


Figure 4: exp66 per-phase runtime (left), peak RAM (centre), and GMRES iteration count (right) versus K . Precompute is K -independent; the block solve is sublinear and the eval linear in K ; peak RAM grows linearly; the block-GMRES iteration count is ~ 30 – 36 (vs. ~ 8 for the single-slab system above).

single resumable job on **10 genoa nodes** (one 96-core worker each; **solve/eval** batches distributed via SlurmManager).

Phase	Wall clock	Detail
prepare	8.80 min	manifest; mostly one-time <code>Pkg.precompile</code>
solve	5.40 h	200 batches, 961.00 s/batch mean (418–1260), dof $\approx 368\,000.00$
consolidate	2.70 min	shared target set (7.40M pts) + ρ -store
eval	26.50 min	200 batches, 78.00 s/batch mean (11–128), 0 hcubature corrections
assemble	40.00 s	gather V (2611^2) + eV tensor
total	6.00 h	10 nodes (≈ 53 node-hours, solve-dominated)

Table 5: Per-phase wall-clock of the 10×10 campaign (job 6525415, 10 genoa nodes). **solve** is $\sim 90\%$ of the wall (200 batches $\times \sim 16$ min on a $\sim 368\,000.00$ -point interface, 10-way parallel).

- **Eval is cheap because there are zero near-singular corrections.** The corrected layer-potential map over the 7.40M-target cloud logged *no* hcubature calculations in any of the 200 batches: the graphene orbital plane (targets at $z \approx 7.5$, support $\approx [4.8, 10.2]$) sits ≈ 1.80 bohr clear of the slab faces ($z = 3, 12$), beyond the near-field range of the target-refined panels, so the map collapses to its smooth FMM far field. The README’s feared eval blow-up does not occur with a thick host slab (cf. exp65’s thin monolayer: $\sim 12,700$ corrections).
- **eV normalization.** $V_{ij,kl} = V^{\text{raw}} 4\pi E_2 / (\|\phi_i\| \|\phi_j\| \|\phi_k\| \|\phi_l\|)$, $E_2 = 14.40 \text{ eV } \text{\AA}$, $\|\phi\|^2 = \int \phi^2$ from the onsite pairs. (Normalizing by the per-pair density norm $\int \rho_{ij}$ instead — the orbital overlap — blows up off-diagonal elements, since it $\rightarrow 0$ for distant pairs.)
- **Quality.** Max relative asymmetry $|V - V^T| / \max |V| = 5.1\%$, dominated by the low-overlap distant pairs (small, noisy V); the dominant onsite/near-neighbour elements are tighter. Tightening `edge_refine_level` ($2 \rightarrow 3$) / `n_quad` would reduce it at higher solve cost.

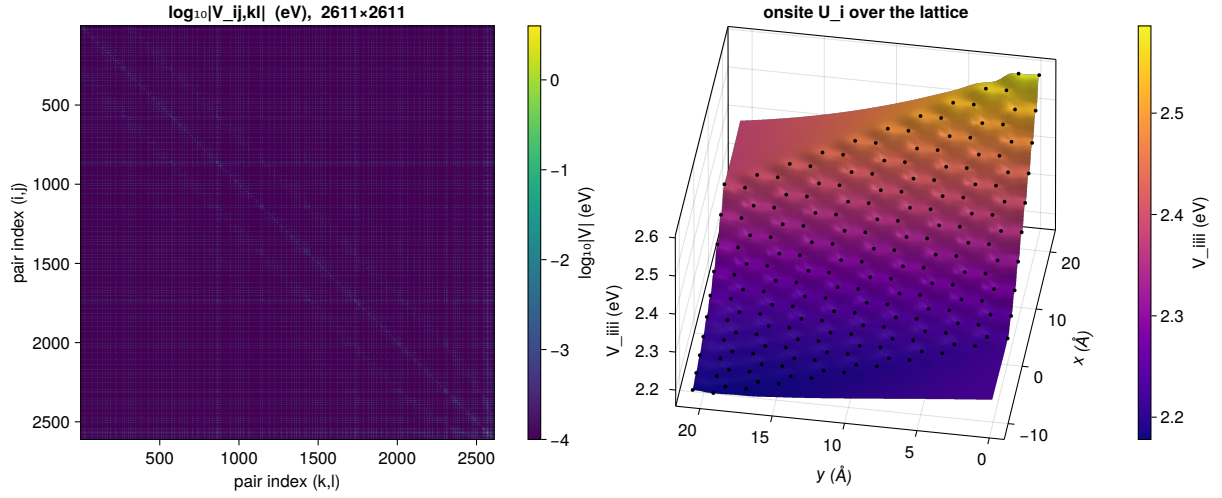


Figure 5: The four-index tensor $V_{ij,kl}$ (eV, orbital-normalized) for the 10×10 lattice. Left: $\log_{10} |V|$ over the full 2611×2611 tensor (pairs in campaign order); the bright diagonal is the pair self-interactions, and off-diagonal couplings fall off 3–4 orders of magnitude. Right: the onsite element V_{iiii} (Hubbard U_i) as a surface over the orbital lattice, IDW-interpolated from the 200 sites (black dots). U_i ranges 2.18 (bulk, maximally screened) to 2.59 eV (flake edge), and is lower over the high- ϵ Si half ($x < 5.5$) than the SiO_2 half — the heterojunction signature.